

09 – Power BI Dashboard and Reports

Architecture slide – Dashboard and Reporting Layer: **Request Status / Cycle Time & Aging / Cost Tracking / Utilization & Performance / Exception Analysis / Executive Reporting.**

1. Connect to data

1. Open **Power BI Desktop** → **Get data** → **Dataverse** → sign in → choose environment **LAM-TVRMS-PROD** (use DEV while building).
2. Select tables: **tvr_tvrequest**, **tvr_feasibilityoption**, **tvr_fundingapproval**, **tvr_milestone**, **tvr_riskissue**, **tvr_costentry**, **tvr_statushistory**, **tvr_department**, **tvr_costcenter**, **tvr_location**, **tvr_researchinstitution**.
3. Mode: **Import** (fast, refresh 4–8 times a day). Use DirectQuery only if live data is a must.
4. In **Power Query**: keep only needed columns; rename them to friendly names; replace choice numbers with labels (the Dataverse connector gives **...name** columns for choices).
5. Add a **Date table** (DAX):

```
Date = ADDCOLUMNS(CALENDAR(DATE(2026,1,1), DATE(2030,12,31)),
  "Year", YEAR([Date]), "Month", FORMAT([Date],"MMM yyyy"),
  "MonthNo", YEAR([Date])*100+MONTH([Date]), "Quarter", "Q" & QUARTER([Date]))
```

6. **Relationships**: TV Request → Department, Cost Center, Location (many-to-one); Feasibility/Funding/Milestone/Risk/Cost/StatusHistory → TV Request (many-to-one); Date[Date] → TV Request[Submitted On (date)].

2. DAX measures (copy into a "_Measures" table)

```
Total Requests      = COUNTROWS('TV Request')
Open Requests       = CALCULATE([Total Requests],
  NOT 'TV Request'[Request Status] IN {"Closed","Cancelled","Rejected","Funding Rejected"})
Closed Requests     = CALCULATE([Total Requests], 'TV Request'[Request Status] = "Closed")
Rejected Requests   = CALCULATE([Total Requests], 'TV Request'[Request Status] IN {"Rejected","Funding Rejected"})
Approval Rate %     = DIVIDE(CALCULATE([Total Requests], NOT ISBLANK('TV Request'[Approved Amount])),
  CALCULATE([Total Requests], 'TV Request'[Request Status] IN {"Funding Approved","Funding Rejected","In Execution","On Hold","Execution Completed","Closed"}))

Avg Cycle Time (days) = AVERAGEX(FILTER('TV Request', NOT ISBLANK('TV Request'[Closed On])),
  DATEDIFF('TV Request'[Submitted On], 'TV Request'[Closed On], DAY))

Age (days) request = DATEDIFF(MAX('TV Request'[Submitted On]), TODAY(), DAY) -- use in table visual per request

Days In Current Status = DATEDIFF(MAX('TV Request'[Status Changed On]), NOW(), DAY)

Approved Budget     = SUM('TV Request'[Approved Amount])
Actual Cost         = SUM('Cost Entry'[Amount])
Cost Variance       = [Approved Budget] - [Actual Cost]
Budget Utilization % = DIVIDE([Actual Cost], [Approved Budget])

On-Time Completion % = DIVIDE(
  CALCULATE([Closed Requests], FILTER('TV Request', 'TV Request'[Closed On] <= 'TV Request'[Required By Date])),
  [Closed Requests])
```

```

Open Issues          = CALCULATE(COUNTROWS('Risk Issue'), 'Risk Issue'[Status] <> "Resolved")
High Severity Issues= CALCULATE([Open Issues], 'Risk Issue'[Severity] = "High")
Overdue Milestones  = CALCULATE(COUNTROWS('Milestone'),
                                'Milestone'[Planned End] < TODAY(), 'Milestone'[Status] <> "Completed")

```

Aging bucket (calculated column on TV Request):

```

Aging Bucket =
VAR d = DATEDIFF('TV Request'[Submitted On], IF(ISBLANK('TV Request'[Closed On]), TODAY(), 'TV
Request'[Closed On]), DAY)
RETURN SWITCH(TRUE(), d <= 15, "0-15 days", d <= 30, "16-30 days", d <= 60, "31-60 days", "60+ days")

```

Time spent in each phase (from Status History) – calculated column on Status History:

```

Hours In Status =
VAR thisReq = 'Status History'[TV Request]
VAR thisTime = 'Status History'[Changed On]
VAR nextTime = CALCULATE(MIN('Status History'[Changed On]),
    FILTER(ALL('Status History'), 'Status History'[TV Request] = thisReq && 'Status
History'[Changed On] > thisTime))
RETURN DATEDIFF(thisTime, IF(ISBLANK(nextTime), NOW(), nextTime), HOUR)

```

Then average of **Hours In Status** by **To Status** = bottleneck chart.

3. Report pages

Page	Visuals	Answers the question
1. Executive Summary	KPI cards (Total, Open, Closed, Avg Cycle Time, Approved Budget, Utilization %), requests by month (column), status funnel	"How are we doing overall?"
2. Request Status	Bar by status, donut by phase, table of open requests with TVID / owner / days in status, slicers (Department, Request Type, Priority, Date)	"Where is every request now?"
3. Cycle Time & Aging	Aging bucket bar, avg cycle time by department, hours in each status (bottleneck), on-time %	"Where do we lose time?"
4. Cost Tracking	Approved vs Actual by request (clustered bar), cost by category (pie), by cost center (matrix), variance	"Are we within budget?"
5. Utilization & Performance	Requests per research institution, avg quoted cost & weeks per institution, budget utilization %, reuse recommendation counts	"Which institutions perform best?"
6. Exception Analysis	Open issues by severity, overdue milestones list, auto-cancelled & rejected requests with reasons, failed notifications	"What is going wrong?"

4. Sample numbers (for testing visuals with sample data)

KPI	Expected with /sample-data
Total Requests	8
Open Requests	5
Closed Requests	1
Rejected / Cancelled	2
Approved Budget	39,500 + 28,000 + 15,000 = 82,500

5. Security in Power BI

- Dataverse connector **respects Dataverse security of the person who refreshes**. With Import mode the service account refreshes → everyone sees everything in the report. So: share the **Executive** report only with APPG + management (`SG-TVRMS-APPG`, leadership group).
- If requesters need a "My requests" report, add **Row-Level Security**: role "Requester" with filter on TV Request: `[Requester Email] = USERPRINCIPALNAME()`.

6. Publish

1. Publish to workspace `LAM TVRMS` (Power BI service).
2. Dataset settings → credentials (OAuth, service account) → **Scheduled refresh** 4× per day.
3. Create an **App** in the workspace and share with the right groups.
4. (Optional) Embed the report in the Canvas app with the *Power BI tile* control, or add it as a dashboard page in the Model-Driven app.